

AI-assisted Co-design of Culturally Engaging Science Stories (ACCESS)

A. PROJECT RATIONALE

This **Integrating Research and Practice Project** is designed to foster science learning and engagement among Latine children ages 3-7 through the development, distribution, and evaluation of a series of 24 dialogic, bilingual e-books centered around culturally engaging narrative science stories. The e-books will incorporate conversational agents allowing children to converse with the characters, as well as family discussion prompts to encourage parents and children to participate in dialogic reading. The books will be co-designed and field-tested with Latine families from community-based early learning organizations in two states, California and Michigan, which will make them more relevant to the range of Latine communities in the United States. Our proposed project is guided by three aims:

Aim 1: *Co-design* 24 Spanish-English bilingual, culturally engaging, and dialogic science narrative e-books with Latine families, leveraging generative AI tools to facilitate more rapid iterations, ensuring that the final products reflect nuanced and authentic family input.

Aim 2: *Examine* the impact of the bilingual science narrative e-books on children's science learning, parents' self-efficacy in science, and parent-child science talk.

Aim 3: *Distribute* the 24 interactive e-books as publicly available content at no cost to children and families through national organizations.

This project will improve young Latine children's informal science learning by creating science resources tailored to their cultural and linguistic assets. Early childhood years are critical for developing the foundational knowledge, skills, and attitudes for later success in STEM (NASEM, 2022). Although the importance of early learning in a stimulating home environment has been widely recognized, access to science learning resources remains disproportionately limited for children and families from minoritized communities (Greenfield et al., 2009). In particular, Latine children represent nearly half of the child population in California and receive less exposure to science starting from early years (Bohlmann & Palacios, 2023; Wonch Hill et al., 2018). Therefore, it is a priority to increase Latine children's access to rich science content.

Furthermore, the available science resources for minority communities are typically designed as culturally neutral. However, science learning is inherently a cultural process (Ladson-Billings, 2014), and the common approach of cultural neutrality often reinforces White cultural dominance, further marginalizing learners from minoritized communities. Thus, it is important to not only focus on increasing the access of science resources, but more importantly critically consider how such access can be grounded in minoritized children's language and cultural practices and lived experiences, which will make science more relatable and encourage children to see themselves as active participants in the field of science (Belgrave et al., 2022; Bermudez et al., 2023). This goal can be achieved by engaging in participatory design processes with community members, which integrates their cultural practices and community values (Bermudez et al., 2023). Latine families have strong storytelling traditions, often passing on STEM content through engaging oral narratives (see, e.g., Castañeda et al., 2022).

Learning from story books is amplified when parents can engage children in interactive exchange of ideas and conversations (often termed as "dialogic reading", Whitehurst, 1992; Whitehurst et al., 1994). The recent advance and prevalence of conversational AI, such as Apple's Siri, Amazon's Alexa, and Google Assistant, offers a new mechanism to facilitate dialogic learning. As a pioneering research group, we have developed conversational agents supporting children's science learning and parent-child bilingual e-book reading. These conversational agents allow children to interact with the media characters in a meaningful and educationally beneficial way (Xu, 2023; Xu, Aubele, et al., 2022) and facilitate the joint engagement of parents (Xu, He, Vigil, et al., 2023).

Despite the potential of conversational media, current development of high-quality educational content and the interactive system for the conversational agents requires extensive effort from content creators, designers, and engineers. This reliance makes the large-scale production of these educational resources challenging. However, this production can be facilitated through the use of generative AI technology, which can automate these tasks (Yu & Guo, 2023) and tailor the content for learners (Su & Yang, 2023). Utilizing generative AI in conjunction with insights from Latine communities surfaced through participatory design provides an avenue to effectively scale the production of customized educational

content while mitigating potential risks of exposing young audiences directly to AI-generated content that might contain misinformation or biases.

In this project we will co-design 24 bilingual science narrative e-books with Latine families to build on their cultural assets to enrich their home environment for science learning. We will also develop a bilingual conversational agent embedded in these e-books to support families' dialogic reading and science learning. In the following sections, we review our on-going NSF grant projects. We then address how this proposal builds on our prior grants and aligns with AISL goals.

A.1 Results from Prior NSF Support

Mark Warschauer (PI), Ying Xu (Co-PI), and Andres Bustamante (Co-PI); award number: 2115382 **Amount:** \$3,000,000 **Dates:** 9/15/2021 – 9/30/2026 **Title:** Developing Conversational Videos to Support Children's STEM Learning and Engagement

Intellectual Merit. This Innovations in Development project is designed to foster STEM learning and engagement among children through the production, evaluation, and distribution of interactive versions of PBS KIDS' STEM television shows incorporating AI-assisted conversation with one of the main characters. The project provides groundbreaking findings on the use of AI-based conversational agents to enhance young children's informal STEM learning. In addition, our project also sheds important light on designing AI-based media that scaffolds children's STEM learning from television. **Broader Impacts.** The final conversational videos will be made available on PBS KIDS' digital platforms that reach 13.6 million unique visitors a month. Data for the study will be collected from low-income Latine families—a fast-growing group whose children face large gaps in STEM outcomes and who spend a great amount of time watching television and video, providing valuable opportunities for STEM learning and interaction.

Publications. This project has resulted in publications in top education and CS venues: Xu et al., 2021; Xu, Vigil, et al., 2022a, 2022b; Xu, Wang, et al., 2022; Xu & Warschauer, 2020; Yao et al., 2022; Zhang et al., 2022. **Products.** The final product of the project will be a web-based application with six conversational episodes for two PBS KIDS shows, *Elinor Wonders Why* and *Lyla in the Loop*, which will be made publicly accessible via PBS KIDS platforms. The source code of this application is shared publicly in the GitHub repository under a Creative Commons license. The conversational script we developed is also downloadable from GitHub. Data for the project is being shared through Dryad, an organization supported by the University of California. **Relationship to Proposed Work.** The proposed project builds on our prior work of developing and evaluating a conversational agent to support children's informal science learning. This agent enhances children's visual and verbal engagement and strengthens parasocial relationships with media characters. Our project advances this innovation by making the conversational agent bilingual in Spanish and English, adding family discussion prompts for parents' involvement, and ensuring its availability on-device for improved accessibility.

Andres Bustamante (PI); award number: 2005776 **Amount:** \$2,570,000 **Dates:** 08/1/2020 – 7/31/2024 **Title:** Stimulating STEM in the City: Co-Designing with Latinx Families to Promote Informal STEM Learning

Intellectual Merit. This project engages community partners in co-design activities to infuse public urban spaces such as local parks, bus stops, and grocery stores with playful informal STEM learning opportunities in low-income Latine neighborhoods. **Broader Impacts.** This project demonstrates new ways to understand the cultural assets that Latine families bring to learning contexts, showing how the unique assets and needs of a local community can be incorporated into public infrastructure.

Publications. Bermudez et al., 2023; Belgrave et al., 2022; Garcia et al., 2022; Chew et al., 2021; Bustamante et al., 2021. **Relationship to Proposed Work.** The proposed project uses similar co-design approaches to develop informal STEM learning resources with Latine families.

Ying Xu (PI); award number: 2302730 **Amount:** \$850,000 **Dates:** 09/01/2023 – 08/31/2026 **Title:** Building a Teacher-AI Collaborative System for Personalized Instruction and Assessment of Comprehension Skills

Intellectual Merit. This proposal is among the first to explicitly focus on scaling up the development of AI-based interactive reading materials to ensure that these resources reach and benefit the broadest possible range of learners. The project will advance state-of-the-art research in automatic question-answer generation and adaptive chatbot systems tailored to educational contexts, which will enable us to

examine a novel mechanism for supporting personalized reading instruction and assessment. **Broader Impacts.** Our research broadens teachers' access to and use of AI by removing the technical burden of creating AI-based materials and directly involving teachers in modifying and refining those materials. This research also broadens children's access to educational AI by enabling the scale up of a publicly available no-cost dialogic reading tool that has been shown in our previous research to have a substantive positive effect on young children's language and literacy development, especially children from marginalized backgrounds. **Publications.** Ziska, Liu, & Xu, 2023. **Products.** The dialogic reading platform for children and teachers to co-create content is available on GitHub. **Relationship to Proposed Work.** This project lays the foundation for testing the feasibility of human-AI co-creating educational media for young children. The method we have developed for training models to generate pedagogical questions can be applied to this proposed work.

B. ALIGNMENT WITH AISL GOALS

Goal 1: Learning STEM in Informal Experiences and Environments

This project builds on our previous work and aims to create a bilingual conversational agent, engage families in science discussions, and distribute dialogic e-books to the public, particularly focusing on the Latine community.

Developing science-themed bilingual e-books with dialogic interactivity. There is a dearth of children's science resources available bilingually in Spanish and English (Bustamante et al., 2018). It is particularly problematic in disciplines like science, where many parents lack confidence due to the content and language barriers which can impede comprehension and learning (Silander et al., 2018). Providing bilingual science learning materials at home also facilitates communication within families which is a rich context for science learning that builds on families' cultural strengths and allows them to leverage their entire linguistic repertoire (Melzi & McWayne, 2023). It also helps children and parents to maintain and develop proficiency in their heritage languages, which is an integral part of cultural identity (Leeman, 2015). In addition, we will develop and evaluate a bilingual conversational agent capable of understanding and producing Spanish, English, and a mix of both, which will improve inclusiveness and learning in Latine children and families. Our exploratory work led to the development of a bilingual conversational agent, which was feasible and well-received by Latine families (Xu, He, Vigil, et al., 2023). This project will represent the first time this technology is brought to scale in a platform with dozens of e-books, and we will evaluate its effectiveness using an experimental design.

Engaging parents and children in science discussions. In addition to involving children in one-on-one dialogues, our bilingual conversational agent will also suggest family discussion prompts for parents and children to have extended discussions about science in the books and in their lives. These discussion prompts will spark science conversation between parents and children (Lemke, 1990), which enhances children's science literacy, competency, and interest (Alexander et al., 2012; Setioko & Ding, 2023). Our pilot studies suggest that family discussion prompts effectively sparked meaningful discussions and enhanced the bonds between parents and children (Xu et al., 2023). This project will further this innovation by incorporating parents' insights into the design of a diverse set of discussion prompts and extend the discussion prompts to topics related to science learning.

Distributing the dialogic e-books to the public. When completed, the dialogic e-books will be distributed as a freely downloadable app in the iOS and Google Play stores. This will facilitate broad public distribution of informal science learning resources tailored to Latine communities and benefit a large number of Latine families (See **Distribution** Section below for more detail).

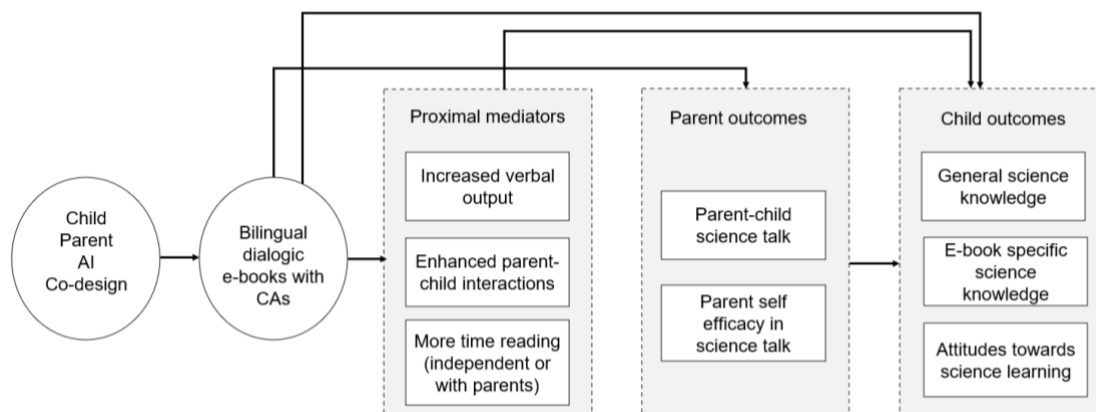
Goal 2: Advancing the Knowledge Base of Informal Science Learning

This proposal will advance the knowledge on scaling up co-designed culturally engaging informal science learning content with AI assistance and enhancing children's informal science learning through AI-powered digital experiences. This project will provide insights into the design, feasibility, and effectiveness of these innovations.

Co-design with AI assistance. This project represents a proof of concept of using AI to scale up informal learning resources created through co-design. We will engage Latine families in the entire process of ideating, developing, and refining the bilingual dialogic e-books. Through participatory design techniques we will surface Latine families' values and home practices and center them in science and dialogic reading experiences. Based on emergent themes from our design sessions, we will investigate

the optimal ways to prompt the AI models to generate narrative stories and dialogic questions that align with families' values and national science learning standards. After generating drafts, we will share the e-books with our co-design family partners and receive feedback to ensure that their voices and values are accurately represented. Subsequently, families will bring the e-books home for one-month of home field testing to gain insights into long-term usability. The co-design and field testing processes will yield valuable insights into designing asset-based informal science materials with and for Latine families.

Examining the effectiveness of dialogic science e-books in informal educational environments. In year 3, we will carry out a one-year home study using a delayed-treatment randomized control trial (RCT) in which families will receive the e-books either the first six months (treatment group) or the last six months of the study (delayed-treatment group). The home-based study will allow us to observe how frequently children and their families voluntarily read the e-books, examine how the interactive features of the e-books affect family interaction during shared reading, understand how family contexts (e.g., parent involvement, technological resources) support or hinder the usefulness of the e-book, and examine science learning that naturally occurs through children's day-to-day interactions with the e-books over an extended period. Using a delayed-treatment study design we will examine to what extent the e-books enhance children's science learning outcomes and attitudes and parents' science self-efficacy and dialogic reading practices in real-life settings. We will also examine the associations between the amount of time families spend using the e-books and the aforementioned outcomes. Additionally, we uphold the ethical standards of the research by guaranteeing that families in the delayed-treatment group will receive the e-books. Since the goal of this project is to disseminate e-books nationwide, the home study will help us identify the strengths and pinpoint common challenges of engaging with AI-assisted conversation in ecologically valid settings, enabling us to make critical revisions before dissemination. A theory of change model is illustrated below:



Goal 3: Equity, Belonging, and Broadening Participation

Using participatory design methodology, this project will co-design dialogic science e-books *with* and *for* Latine families by **leveraging funds of knowledge and existing cultural practices**. The cultural values and funds of knowledge that Latine families bring to bear, especially the tradition in oral storytelling (Delgado-Gaitán, 2019; Sánchez et al., 2010), shape our research. Various forms of oral discourse, such as family reminiscing about past shared events; traditional stories highlighted by *dichos* (common sayings); and concepts like *consejos* (advice), *refranes* (proverbs), and *adivinanzas* (riddles) are employed by Latine families to facilitate children's learning (Melzi et al., 2019). Additionally, Latine families often engage in STEM-related oral storytelling and conversations about animals, plants, astronomy, and weather (Callanan et al., 2017; Castañeda et al., 2022; Kelemen et al., 2014; Perez-Granados & Callanan, 1997; Shirefley et al., 2020). There are also abundant science learning opportunities in families' everyday routines like cooking, gardening, laundry, or grocery shopping. Therefore, adopting a narrative format of the science learning content is deeply rooted in Latine families' cultural practices and can connect to their everyday routines. In addition, families will be engaged in the

entire development phase, including the ideation, design, and piloting of the e-books, to ensure the books reflect their voices, sustain their cultural values, and are usable in their home environment.

Goal 4: Intentionally Community/Practitioner Driven

The development of the science dialogic reading materials will be underpinned by community involvement, with families and children serving as our design partners. Their stories and perspectives will be centered in the materials. We will conduct this work in close collaboration with two community organizations: the Santa Ana Early Learning Initiative (SAELI), a non-profit organization in California that serves more than 250 children and families, and En Nuestra Lengua (ENL), a non-profit based in Ann Arbor, Michigan, which serves over 40 children aged 4-5 and their families. These community organizations will facilitate the identification and recruitment of local families who will be positioned as experts in their cultural practices and routines and will collaborate in infusing those values and practices into early science materials. Furthermore, our partnership with families from two distinct Latine communities acknowledges the diversity within this cultural group, enhancing the relevance and relatability of our results for national distribution. While SAELI families are predominantly of Mexican origin, families in ENL come from every Spanish-speaking country in the continent except for Paraguay.

B.1 Broader Impacts

The project will directly involve 160 Latine children and their families in Southern California and Michigan through participation of two of the largest and most influential Latine-led early learning organizations in those states. Thousands more Latine families will be reached in the final year of the project through ambitious national distribution of the free e-books with the assistance of the Spanish/Latino Culture editor of Common Sense Media and additional project advisors from PBS KIDS, Sesame Workshop, and Age of Learning. Three early career researchers—all from immigrant backgrounds—will participate in the leadership team, and graduate and undergraduate students at both UCI and UM with requisite cultural and linguistic expertise to conduct research in Latine immigrant communities will be recruited and receive research training. Results from the project will be broadly disseminated to support further use of AI to develop culturally engaging STEM learning experiences, and will also be incorporated into undergraduate, teacher preparation, and graduate training at UCI and UM.

C. PROJECT DESIGN

C.1 Research Foci

The main objective of this project is to co-design, evaluate, and distribute bilingual interactive science narrative e-books. The sections below detail our research plan.

C.2 Methods and Analyses

We will use a design-based research approach in which we co-design, develop, pilot, refine, and evaluate 24 interactive e-books to see how they function in informal educational environments, with the end goal of distributing scalable and effective science digital products for real-world usage.

C.2.1 Co-design of the Bilingual Interactive E-books

Co-designing 24 Science Narratives and Dialogic Reading Experiences

In the first two years, we will iteratively co-design 24 science narratives and the accompanying dialogic reading experiences with parents from SAELI and ENL. This e-book series will feature a curious Latine girl who loves to ask questions and find answers using scientific inquiry with her friends and family members at her home and in community spaces.

The overall structure of resources will be guided by the three dimensions of K-12 science learning suggested by the Next Generation Science Standards (NGSS; National Research Council, 2013). The 24 books will consist of six books in each of the four core science domains: life sciences, physical sciences, earth and space sciences, and engineering and technology. The science and engineering practices and crosscutting concepts will be integrated into all the narratives.

Co-design cycles will be conducted simultaneously across two sites (SAELI and ENL) during the first two years of the project. Each site will develop 6 narratives and dialogic experiences for one domain per year. In each co-design cycle, 10 families will convene monthly for two hours with the research team, yielding 12 co-design sessions for each set of books. The sessions will be conducted in Spanish (translation will be available for research members who do not speak Spanish).

At the beginning of each co-design cycle, we will present the goals for the project to parents to orient them to the plans for the year. Next, we will conduct storytelling activities to elicit families' values, assets, and practices related to informal science learning (Pstross et al., 2017). Subsequent sessions will include a variety of other design techniques such as storyboarding, bags of stuff, madlibs, or photo elicitation (Bermudez et al., 2023). For example, we might encourage them to share pictures of the places they like to visit with their children and explore the science learning that takes place in those spaces (Le Dantec et al., 2009). Through these co-design activities, we will identify the science learning opportunities situated in their cultural practices, and inspire potential ideas and themes for the narratives.

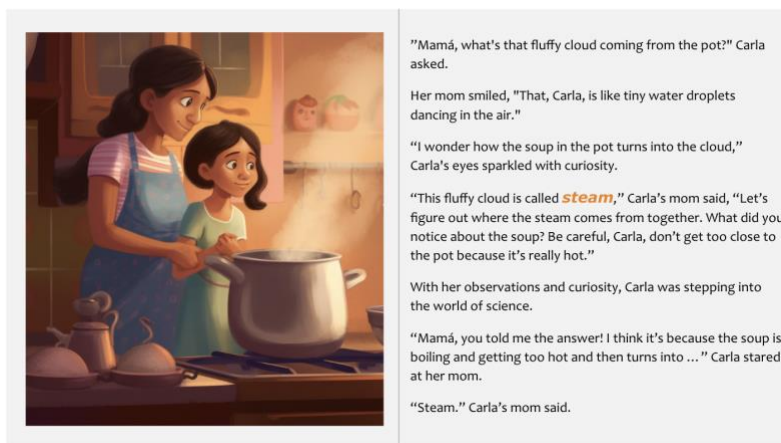
Next, the research team, with consultancy from experts in child media and science learning, will align the identified narrative themes with the NGSS framework, emphasizing connections to the disciplinary core ideas, science practices, and crosscutting concepts. For example, during a co-design session in our prior NSF project, a parent shared a story about her child helping her cook Caldo de Res and noticing water in the soup turns into steam. This very common and familiar life experience opens opportunities for families to discuss the *Disciplinary Core Idea* of physical science (the states of matter can change under different temperatures), engage in the *Science Practices* of asking questions and constructing explanations, and notice the *Crosscutting Concepts* of cause and effect relationships or stability and change (heating water causes it to change into vapor). All of a sudden this everyday practice is transformed into a rich science learning experience covering all domains of the NGSS framework.

Simultaneously, we will investigate the optimal ways to generate narratives by using generative AI tools such as GPT-4 and Gemini. We will fine-tune a large language model (such as GPT-4 and Gemini) using science narrative texts from reliable sources, such as science trade books that have won awards from the National Science Teaching Association (NSTA) in the past decades. In addition, we will conduct prompt engineering to identify the optimal approaches to prompt the model so that it generates a more engaging, coherent, educational, and developmentally appropriate story draft. The story generation will be an iterative prompt and re-prompt process until the research team considers it suitable to proceed to the next phase. The research team will then discuss and revise the narratives internally and develop the English version (original drafts will be written in Spanish) of the narratives before the team brings the refined bilingual drafts to the consultancy panel for further discussion and final revision. Throughout this interactive process, we will consider text complexity/difficulty using quantitative indicators (e.g., Lexile and Coh-Metrix) and qualitative measures (e.g., text structure, knowledge demands), as well as ensure an authentic representation

of the Latine communities' values, norms, and practices. A sample page of a story was generated and refined using the aforementioned techniques based on parental input presented in the right figure.

After crafting the narrative drafts, we will present them to the parents and showcase how their insights during the earlier co-design sessions contributed to these narratives. We will give time for the parents to read and discuss the narratives as a small group and elicit feedback and suggestions regarding these narratives' readability, coherence, cultural representations, educational values, etc. Additionally, we will swap the narrative drafts between the two co-design groups to obtain diverse perspectives and feedback and build mutual understanding across sites.

After we craft the narratives, we will design the dialogic reading questions and prompts with parents. We will first present to the parents the dialogic reading features accompanying these e-books. Next, we will ask parents to share their routines and experiences around book reading, both in printed and digital formats. On this basis, we will brainstorm with parents the kinds of questions they will ask and interactions



they will have with their children when reading these science narratives to fulfill their different goals. These stories will reveal families' cultural practices and book-reading routines that can be integrated into the design of the dialogic reading experiences for the science e-books.

Meanwhile, we will have parents bring these stories home and read with their children, followed by co-design activities in which parents form small groups and share their co-reading experiences, including the occasions in which they read with their children, their interactions with their children, whether their children like the stories, what learning opportunities they see in the stories, etc. We will also collect the kinds of questions parents asked and the discussions they had with their children during the co-reading. These sessions can further inform the dialogic questions and surface the natural discussions families have when reading with their children. The findings will form the basis of the development of the conversational script for the bilingual conversational agent that is depicted in the next section.

Building the Bilingual Conversational Agent and E-books

The e-book for each story will allow the users to choose the language of the print text and audio narration, either Spanish or English. The e-books will by default play the audio narration page-by-page, but the parents and children will have the option to turn the narration off if they wish to read on their own. Children can choose to read alone or with their parents. In both the English and Spanish versions, the embedded bilingual CA will be capable of comprehending and carrying out dialogue in both languages (i.e., children may use English, Spanish, or a mixture of both when interacting with the CA, and the CA will reply in either English or Spanish depending on the language the child principally uses).

The conversational script for the bilingual conversational agent will be developed based on parental insights gained from co-design activities and NGSS. Two forms of dialogue components will be included in the e-books: direct conversation between the conversational agent and the child, and family discussion prompts in the middle and at the end of the books to stimulate family discussion.

In the agent-child direct conversation, the conversational agent will ask children questions at the end of each page, listen to and interpret their answers using natural language processing, and then provide responsive feedback based on the children's answers. The main learning goal of this dialogue component is to strengthen children's understanding of specific science vocabulary, concepts, and practices. The family discussion prompts will seek to extend the preceding child-agent conversation by providing the child an opportunity to relate the story to their own science-related experience and have an extended discussion with their parents. Such prompts not only actively engage parents in the co-reading, which benefits children's learning, but also foster children to transfer what they learn from the e-books to their daily science experiences.

Accompanying the narratives on water evaporation, for example, multiple dialogic questions and family discussion prompts can be constructed. For instance, the character can prompt the child to investigate by asking "What do you notice about the soup in the pot?" After the child understands that heat causes water to evaporate and become steam, the character can probe, "What if we turn off the stove and let the soup cool down? Let's make a prediction. A prediction is when we guess what is going to happen. What do you think will happen to the steam if we turn off the stove?" Moreover, the character can suggest a family discussion prompt, "It's time for a family discussion. What other changes happen to water in very hot or very cold temperatures? And where have you seen these changes before?", which connects the topic of different states of water to their daily life. These dialogic questions and discussion prompts emphasize one or more Science Practices, Crosscutting Concepts, and Disciplinary Core Ideas.

The alignment between the sample narrative, dialogic questions, and the NGSS framework:

Narrative: The character is helping her mom cook Caldo de Res. She notices the fluffy cloud rising from the pot into the air and wonders what that is. With support from her mom, the character makes observations of the boiling soup with bubbles popping up on the surface and forms the hypothesis that heat causes the water to become steam. Her mom then elaborates and explains the process of evaporation.

Dialogic Questions and Discussion Prompts	Science Practices	Crosscutting Concepts	Disciplinary Core Ideas Physical Sciences - Matter and Its Interactions:
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Let's make observations together! What do you notice about the soup in the pot?	Making observations	System and system model Stability and change	Observe and describe characteristics of objects
Why does the soup turn into steam?	Constructing explanations Engaging in argument from evidence	Cause and effect Stability and change	Describe the states, properties, and reactions of matter
What if we turn off the stove and let the soup cool down? What will happen?	Developing and using models	Patterns Cause and effect Stability and change	Predict the states of matter
<i>(Discussion prompt)</i> It's time for a family discussion. What other changes happen to water in very hot or very cold temperatures? And where have you seen these changes before?	Developing and using models	Patterns	Predict the states of matter

In CA-child direct conversation, the CA will provide responsive feedback based on a child's answer. The CA will analyze children's answers using GPT-4 and perform end-to-end processing to first transcribe speech into text and then classify the transcription by semantic intent. Given that children can respond to a particular question in a variety of ways, the agent will be trained to associate more than one semantic intent with each conversational opportunity. These intents will be created based on predicted responses formulated by the research team and based on children's utterances during field testing.

We will also include a fallback intent that is triggered when a child's utterance cannot be classified into any of the predefined intents or when the child does not respond to the question at all. When a fallback intent is triggered, the agent scaffolds the conversation by rephrasing the original question using more accessible language (e.g., changing from an open-ended question to a multiple-choice question). If the child's response to the scaffolded question triggers the fallback intent again, the agent then provides neutral feedback and explains the correct answer to the child. This scaffolding and fallback mechanism is important for extending the conversation while preventing communication breakdowns and frustration, and we have successfully implemented this in our previous work (Xu & Warschauer, 2020).

We will create NLU modules capable of processing both English and Spanish responses. Specifically, we will first deploy a multilingual detection function provided by Google that enables the CA to detect whether an utterance is in Spanish or English, and then process this utterance using the appropriate NLU module. If a child responds using both languages within a single utterance, the CA will process the utterance through both the English and Spanish NLU modules simultaneously and then respond based on the intent returned by whichever module most confidently classified the utterance.

We have already created and piloted a bilingual CA to confirm its feasibility. We tested this CA's performance using speech samples collected from 20 children proficient in Spanish and English, who answered dialogic questions while listening to the story *Rosita and Elmo Read a Recipe* in both Spanish and English. Within these speech samples, 27% of the utterances were labeled as between-sentence code-switching and 6% were within-sentence code-switching. The CA successfully interpreted 91% of the audio samples and subsequently classified them into the corresponding intent. This performance was consistent with that of the monolingual CAs we deployed in our prior studies (Xu, Aubele, et al., 2022; Xu et al., 2021; Xu, Vigil, et al., 2022b), indicating the technical feasibility of a bilingual CA in this context.

The family discussion prompts do not involve automatic responses from the CA. Instead, after posing the prompt, the CA allows parents to guide their child in discussion on the topic. Current conversational

technology is not sufficiently capable of reliably processing speech input from multiple users. Nevertheless, this design has the benefit of allowing more flexibility for richer parent-child discussions while also encouraging parents to take a more active role in learning-focused dialogue during storybook reading with their child (Troseth et al., 2020; Yang et al., 2022).

Piloting the Interactive E-books at Homes

At the end of each co-design cycle, we will conduct a one-month home-based pilot study with involved families. The goal of the home-based piloting is to examine the feasibility and families' perception of dialogic e-books in ecologically valid settings. Families will be provided with tablets installed with the e-books and consent to automatic voice recording during e-book reading. We will request they read the 6 e-books at least once and encourage them to read as many times as possible. This ensures that we obtain data and feedback from every e-book. Additionally, families will be able to make their own choices regarding the e-books' language. During the piloting, the research team will be available to assist via text, phone call, or video conference if the families encounter any technical difficulties. Weekly surveys will be distributed to each family to understand their use of the e-books. Log data will be stored and analyzed to understand the performance of the bilingual conversational agent. Parent and child interviews will be conducted at the end of the piloting to understand their perceptions of the e-books, feedback, and suggestions for further improvement.

C.2.2 Evaluation of the Effectiveness Through a Home-Based RCT

The evaluation of the bilingual interactive science e-books include a six-month home-based study to determine the effectiveness of these e-books for science learning, to understand how children and families engage with these e-books, and how learning occurs in a natural setting.

Utilizing the delayed-treatment randomized control trial (RCT) study design, we will randomly assign each family to the treatment or delayed-treatment condition. Half of the families will be given the e-books at the start of the study and the remaining half in the delayed-treatment condition will not be provided with the e-books until six months later. Families in the treatment condition will receive a web app containing 24 bilingual interactive e-books. To allow us to understand naturalistic use and evaluate the agent's on-device performance, families will use their own devices (smartphones, tablets, or computers with cameras and microphones, per their choice) and consent to automatic voice recording during e-book reading. We will request that they read at least one new e-book each week, so that they read all 24 e-books in the study period. Families will make daily choices in terms of the e-books' language and whether to read together. This flexibility will allow us to preserve the ecological validity of our findings. All study sessions will take place at participants' homes.

To facilitate families' e-book usage, the research team will provide instructions to the participants on how to operate the tablet and e-book app. At the start of the home-based study, researchers will demonstrate how to download the e-books, the e-book's functions and then encourage the parents and children to try reading a page together. During the home usage, the team will also be available to assist via text, phone call, or video conference should participants encounter technical difficulties or have questions. Log data will be used to monitor participants' use of e-books, including whether and how many of the books are read, the time and duration, the language they choose to read in (Spanish or English), and the person who reads with the child. If the team finds that participants are not reading the books, they will follow up with the participants to remind them to do so. In addition, the device's built-in microphone will be activated once the participants open the e-book app to audio record the speech generated by children and parents. We will implement a child-friendly passcode for login with a drop-down menu listing all consenting family members to identify who is reading the e-book, including any caregivers or children (the participating child or siblings). We have tested this approach in our feasibility study and the research team was able to utilize audio recordings to transcribe conversations during the reading process.

Participants and Sample

Participants for the co-design phase will be recruited from Santa Ana, California and Ann Arbor, Michigan. Twenty Latine parents of four-to-seven-year-olds will be recruited from SAELI in Santa Ana and En Nuestra Lengua in Ann Arbor. SAELI is a community organization that serves Latine families in Santa Ana and that we have conducted numerous co-design projects with. It was the 2023 winner of the John and Louise Byrson Transformative Leader in Early Childhood Award from PBS SoCal. En Nuestra Lengua is a Saturday Spanish immersion program for Spanish heritage speakers, ages 4 - 10. Their Pre-

K and K curriculum is centered on "Proyecto WORLD," a home-based dialogic Spanish reading intervention to promote vocabulary expansion, particularly of scientific (age-appropriate) terms.

One hundred and twenty Latine families will be recruited for the home-based RCT, with half in the treatment group and half in the delayed-treatment condition. Eighty will be recruited from Santa Ana and forty will be from ENL.

For the home-based RCT, we anticipate an attrition rate of 10% based on similar home intervention studies. With this assumption, we estimate 110 families will complete the study, resulting in 55 families each in the treatment and delayed-treatment conditions. If not considering the inclusion of baseline and child pretest science knowledge as covariates in our analysis, a sample size of 55 in each condition will provide 80% statistical power to detect a minimum effect size of .27 (Cohen's f) at $p < .05$ in an ANOVA analysis on child outcomes. Using ANCOVA with pretests included as covariates (e.g., children's age, language proficiency, prior science knowledge) that are expected to be strongly correlated with the outcome variables, we expect that the minimum detectable effect size will be reduced to below .20. Our research indicates that conversational-based learning yields a medium effect size on both children's story-specific reading assessment and episode-specific science knowledge. Specifically, we observed effect sizes ranging from $f = .21$ to $f = .31$ for reading assessments (Xu, Aubele, et al., 2022; Xu et al., 2021), and from $f = .32$ to $f = .42$ for science knowledge assessments (Xu, He, Levine, et al., 2023; Xu, Vigil, et al., 2022a). Prior research on dialogic reading home interventions typically suggests a small to medium effect size on children's standard language outcomes (Mol et al., 2008). Thus, a sample size of 120 for the RCT is sufficient to detect the expected effect.

Data Sources

We utilize a battery of testing and observation to measure baseline information, child and parent outcomes, as well as within- and across-session language and engagement behaviors.

Baseline data. Baseline data will be collected prior to the home study.

Children's demographic information will be collected through a parent survey, in which parents report their child's biological sex, birth date, race/ethnicity, parent's education level, income, home language environment, etc.

Children's language proficiency in English and Spanish will be assessed via the Quick Interactive Language Screener: English–Spanish (QUILS: ES; Levine et al., 2020). The QUILS: ES comprises three short tests (vocabulary, syntax, and process) for each language. Each test contains 14–16 items that are sensitive to development. A score for each skill type across both languages will be computed.

Pretest and posttest measures will be collected both prior to and after the home study for both children and parents.

Children's general science knowledge will be assessed using Lens on Science (Greenfield, 2015), a computer-adaptive assessment targeting preschool-aged children to measure children's baseline science knowledge. It was developed according to the National Research Council's Framework for K-12 Science Education. Lens assesses children's knowledge of three science areas, life science, earth/space science, and physical/energy science, and children's science processing skills, and consists of 35-40 items in total.

Children's science knowledge of the e-book content will be assessed via researcher-developed questionnaires pertaining to specific e-books. Specifically, these items not only focus on assessing children's understanding of science vocabulary, concepts, and practices presented in the e-books, but also aim to evaluate whether they are able to transfer and apply the science inquiry practices to other novel scenarios. The assessment items will be based on the NGSS framework. The items will be piloted in the field-testing stage with at least 20 children to ensure their reliability and validity. We have used a similar approach to developing an assessment questionnaire in our prior NSF grant projects (Xu et al., 2022). The Cronbach's alpha internal consistency of that assessment questionnaire is 0.73 among 11 items, and the Pearson correlation between children's baseline language score and the assessment is 0.56, indicating good concurrent validity.

Children's attitudes towards STEM learning will be measured in the home study. We will use the instrument "Puppet Interview Scales of Competence in and Enjoyment of Science (PISCES)" developed by Mantzicopoulos et al. (2008). The scale consists of 16 items with a dichotomous response format along three dimensions: science competence (e.g., "I know a lot about different kinds of living things"), science liking (e.g., "I like using different science tools"), and ease of science learning (e.g., "I can figure things out in science"). The examiner will use two gender-neutral puppets throughout the administration of

the scale to facilitate responses. The procedure will involve one puppet making a statement representing the positive end of the continuum for an item (e.g., “Science is easy for me”) followed by a statement from the other puppet representing the negative end of the continuum for the same item (“Science is hard for me”). To control for order effects, the positive statements made by each puppet will be counterbalanced so that each puppet makes both positive and negative statements. The test will last about 15 minutes. The Cronbach's alpha internal consistency of this scale is 0.84 as reported in Mantzicopoulos et al. (2008).

Parents' self-efficacy/beliefs in STEM. Parents' attitudes and beliefs towards science will be assessed using an adapted version of the Preschool Teachers Attitudes and Beliefs Towards Science (P-TABS; Maier et al., 2013) questionnaire. P-TABS uses a 5-point Likert scale and has demonstrated reliability and validity in Head Start centers. Items include concepts around comfort with science (e.g., “I am comfortable doing science activities”) and the perceived usefulness and appropriateness of science in early childhood (e.g., “Science activities are too difficult for children” or “Young children are curious about science”). This measure has been adapted for use by parents in previous studies by our research team (Bustamante et al., 2020).

Parent-child reading behaviors on science narratives will be captured by a shared reading task, in which participants will be given a different science narrative book in the pretest or posttest, and read in either English or Spanish as they prefer. We will counterbalance which story participants receive for the pre and posttest. The session will be recorded for analysis of parent-child interactions in two folds. First, we will capture the nature of parent-child science talk by adopting a coding scheme developed by Tenenbaum et al. (2005) focused on science process talk, descriptive properties, task-related talk, conversational talk, etc. We will also measure the quality of verbal exchanges between parents and children using a 12-item scale, a subset of the parent-child interaction coding scale developed by Adamson et al. (2012). The scale contains three subscales: the child, the parent, and the shared topic. The child subscale contains four items: initiation, responsiveness, expression, and affective communication. The parent subscale contains four items: scaffolding, symbol highlighting, following in on the child's focus, and affective communication. The shared topic subscale contains four items: elaboration, sustainability, scope, and fluency and connectedness of conversation. Each item utilizes a 7-point Likert scale.

Mechanism measures. Interactions during e-book reading will be audio recorded and scored for analysis of the reading process, using validated coding protocols. Log data will also be obtained during the home use of e-books.

Child and parent verbal output will be captured via consented voice recording and sharing. We will measure the length (total number of words), lexical diversity (total number of unique words), topical relevance (whether a child directly answers a question), accuracy (whether a child's answer is correct), or intelligibility of children's utterances (whether a child's answer is intelligible). For details of these measures, please see our earlier paper (Xu, He, Levine, et al., 2023; Xu & Warschauer, 2020).

Parent-child interaction will be analyzed with audio recordings in terms of science-related talk and quality of verbal exchanges, using the same coding scheme as our pre and post measures of parent-child reading behaviors on science narratives (see Parent-child reading behaviors on science narratives).

Log data. The performance of the conversational agent's language models will be analyzed via a comparison between (1) the dialogue log and (2) the transcription of each child's utterances by a human researcher. To determine the accuracy of speech recognition, we will compare the records of the conversational agent's recognition of speech in text form with the related human transcription. To determine the accuracy of intent detection, we will compare intents detected by the conversational agent with those classified by human researchers. A score of intent detection confidence retrieved from log data will also be used to evaluate the natural language training models. We will also collect additional log data on families' usage of the e-books to evaluate families' frequency and duration of video watching and to understand families' choice of e-books and e-book languages (English or Spanish).

Analyses

Aim 1: The main goal of the **development phase** is to co-design bilingual dialogic e-books that meet the needs and expectations of Latine families and align with the NGSS framework. In this phase, we will primarily utilize qualitative methods and descriptive statistics for data analysis. Two key sources of data will be audio/video recordings of 1) the co-design sessions where parents engage in design

activities, and share their perceptions and feedback on the e-books and 2) participants' interactions with the research products at various stages (e.g., stories, conversational prompts, e-book app). These data will be transcribed and combined with the study session field notes to conduct a thematic analysis. We will document emergent themes around family cultural values and practices through an inductive coding process and identify patterns in how participants engage and interact with the e-books. During the one-month home-based piloting of the e-books, log data will be analyzed to assess the performance of the natural language processing models.

Aim 2: The main goal of the RCT is to estimate the effects of dialogic science narrative e-books in naturalistic home settings on children's science learning, and parent-child interaction outcomes, and to examine heterogeneous effects depending on children's and parents' characteristics. We will also examine whether and how dialogic science e-books affect children's science learning through impacting child engagement in reading science content, and through impacting parent dialogic reading behaviors and science-related discussions with children.

To examine the main effects, we will use regression analysis to estimate the condition effects, with the child as the unit of analysis. In addition, we will conduct exploratory analyses on the heterogeneous effects of the dialogic science e-book treatment based on children's pretest measures (i.e., bilingual proficiency and standard and book-specific science knowledge). We will also conduct another set of exploratory analyses to examine whether the dialogic science e-book treatment has a causal effect on mediating factors that lead to child outcomes. Structural equation modeling (SEM) will be conducted to test the mediating hypotheses with the assigned treatment conditions entered as a multi-categorical predictor. The hypothesized mediators include children's and parents' reading behaviors.

C.2.3 Distribution to a Broad Audience

In the final stage of the project, we will develop and distribute a mobile iOS and Android app hosting the 24 dialogic science e-books. Instead of the *online* conversational agent used in earlier stages of the project, the mobile app will deploy *on-device* speech recognition and intent classification, using approaches we have developed and proven feasible in our current AISL-funded project on dialogic videos. This will eliminate any API fees that online engines charge, better protect children's privacy, and make the apps usable without an Internet connection.

The mobile app will be made available at no cost on the iOS App Store and Google Play store. PBS KIDS will include the e-book description and links on their Parent Resources and Tips page. The Joan Ganz Cooney Center at Sesame Workshop will feature a blog about the app. The Latino Culture page at Common Sense Media will feature a review of it. AB member Aviles will assist in reaching out to major organizations she has worked that reach a large number of Latine families, including the Univision Spanish-language media network and First 5 California. Co-PIs Bustamante, Satterfield, and Soto-Lara will coordinate with organizations they participate in, including the Hispanic Research Center, the National Centers for Families Learning, the Latino Family Literacy Project, Empowering Latino Futures, and the Boys & Girls Clubs of America. A small budget has been set aside for use of these organizations' email lists or other distribution channels such as Website announcements.

Our research team will measure the distribution success and user engagement by tracking the number of downloads from the App Store. Furthermore, new users will be asked to fill out a short registration form to collect demographic information, including geographical location, the age of their child, parental education level, and home language. This data will help us assess whether our e-books are effectively reaching our intended audience.

C.3 Plan for Implementation

The timeline for co-design, evaluation, and distribution is detailed below.

	Year 1				Year 2				Year 3				Year 4			
Quarter	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
Co-design 6 books in domain 1 (CA)																

Co-design 6 books in domain 2 (MI)																	
Co-design 6 books in domain 3 (CA)																	
Co-design 6 books in domain 4 (MI)																	
Home study of 24 books (CA & MI)																	
National dissemination																	

Note: Domains 1-4 refer to four domains covered by the Disciplinary Core Ideas suggested by NGSS, including life science, physical science, earth and space science, and engineering and technology.

D. COMMUNICATION PLAN

Our communication plan will focus on five audiences: scholars, educational media producers, informal educators, educational organizations, and families. The purpose of the communication will be to disseminate both our findings (via publications, presentations, data sets, and popular summaries) and our products (our 24 e-books).

Scholars will be reached through publications and presentations in academic outlets related to science learning, digital learning, early childhood development, and bilingual education. PI Warschauer is one of the most widely cited researchers on educational technology in the world. His Digital Learning Lab website, which lists papers, presentations, and reports from his research projects, also receives tens of thousands of unique visitors a year. Venues for scholarly publications include *AERA*, *Child Development*, *CHI*, *Computers & Education*, *International Journal of Bilingual Education & Bilingualism*, *Learning & Instruction*, *Journal of Applied Developmental Psychology*, *Interaction Design & Children*, *Association for Computational Linguistics*, *Journal of the Learning Sciences*, *Early Childhood Research Quarterly*, and *Learning and Individual Differences*. All data from the project will be published in easily accessible archives, further assisting scholars engaged in replication or related work. We will also share information with research associations relevant to the topic, including the National Research Center on Hispanic Children & Families and Connecting the EdTech Research EcoSystem (CERES), both of which we have close relationships with.

For the last several years, we have had meetings on a regular basis with several of the largest **educational media producers** in the country including PBS/PBS KIDS, the Corporation for Public Broadcasting, Sesame Workshop, and Age of Learning. We will continue those meetings and share the results of our project.

We will reach out to **informal educators** through presentations, meetings, and written dissemination to the community organizations mentioned in the dissemination plan above as well as to the National Informal STEM Education Network and Reimagining Equity and Values in Informal STEM Education. We will also send information out through the Connected Learning Alliance, with which we are affiliated, and present at its Connected Learning Summit, which reaches many informal educators.

As for **educational organizations**, we will make presentations to the Santa Ana Unified School District, Montebello Unified School District, and El Sol Science & Arts Academy—three local educational agencies serving 95%+ Latine students and with whom we have research-practice partnerships on other initiatives. We will also have contacts in all County Offices of Education in California and will forward them information and invite them to a video conference presentation on our findings. Over the past year, we have made presentations on AI in education to County Offices of Education in Los Angeles, Orange, Ventura, and Santa Clara counties, and we will reach out to them to present again. California has recently passed legislation mandating that public school districts provide transitional kindergarten enrollment to all 4-year-olds in the state, but there is little educational material available for this, so we anticipate there will be great interest in our e-books. Finally, we will distribute information about the project through the Orange County Educational Advancement Network, a collection of schools, districts, and non-profits in the county led by the UC Irvine School of Education (Co-PI Bustamante is the incoming director).

Our e-books will be made available to **families** across the country. As the e-books will be designed for on-device use, they will require no Internet access to use, and there are no risks of online access to children's data. The e-books will be made available for free download on both the iOS (for iPhone and iPad) and Google Play (for Android phones, Android tablets, and Chromebooks) stores. In all the communications described above, we will publicize the availability of the e-books and ask partners to communicate with them. In addition, Warschauer and Xu have previously published multiple essays in *The Conversation*, a website of public commentary by academics that reaches millions of readers a year; they will seek to do so again. Warschauer's prior research has also been published in *Edutopia* and *EdSurge*, publications that similarly reach large numbers of researchers, and the team will reach out to share their information. Information will also be shared with The Academic Minute podcast and Spectrum News broadcast, both of which have publicized Warschauer's projects previously. These venues will be used to share information about the project as well as the availability of the downloadable materials. Co-PI Bustamante has blogged previously for *Huffington Post*, Brookings Institution, and BOLD Blog, and will blog widely on this project.

Finally, the **project website** will be designed to facilitate communication with all these audiences. The website will be part of the UCI Digital Learning Lab website, which receives tens of thousands of visits a year, and will also be linked to Bustamante's and Xu's lab websites. It will include an overview of sample videos of children using the e-books (with consent of parents); links to presentations, reports, data sets, blogs, and essays; videos about the project; and links to the download sites on the iOS and Google Play stores. The School of Education at UCI and the Marsal Family School of Education at UM will distribute information about the project in their newsletters, magazines, and distributional mailings reaching tens of thousands of alumni, donors, and educators across the country.

E. EVALUATION

Evaluation will be carried out with the assistance of the Advisory Board (AB) of international experts in the critical areas of the grant (i.e., informal science learning among children, co-design with Latine communities, learning through conversational e-books). We first introduce the Advisory Board members and then discuss the role of the Board in evaluation.

E.1 Advisory Board

Maria Aviles is Spanish editor at Common Sense Media, where she serves as both reviewer of all Spanish language material as well as the point of contact for media companies conducting outreach to Latine communities. She has extensive networks with educational and media companies serving the Latine community, such as Univision. **Susana Beltran-Grimm** is a Postdoctoral Research Associate in Human Development and Family Science at Purdue University. Her research focuses on co-design of children's media for STEM learning with Latine families so as to build from families' values, practices, and funds of knowledge. **Robert Kalinowski** is Director of Design Research and Consumer Insights at Age of Learning, with expertise in design of children's STEM and literacy materials that are both engaging and educational. **Silvia Lovato** is Senior Director and Head of Learning and Research at PBS KIDS with expertise on children's learning from interactive media. **Gigliana Melzi** is a Professor of Applied Psychology at New York University. Her work focuses on the intersection of Latine families' cultural and linguistic practices around stories and their relation to children's early development and learning in STEM. **Medha Tare** is Senior Director of Research at the Joan Ganz Cooney Center at Sesame Workshop, with expertise in children's acquisition of language, social, and conceptual knowledge through interactions with parents or in informal contexts such as museums and media. **Georgene Troseth** is Professor of Psychology, Emerita, at Vanderbilt University. Her research focuses on the use of AI-based e-books to support dialogic reading in families. **Daniel Whiteson** is Professor of Physics and Astronomy at UCI and co-founder of the children's media company, Shoe Ink. He has extensive experience developing narrative stories to promote science learning for young children, including as developer of the Elinor Wonders Why children's television show on PBS KIDS.

E.2 Evaluation Approach

The Advisory Board will be consulted regularly both through scheduled quarterly meetings as well as interim consultations with Board members with particular expertise. We have found in previous AISL-funded projects that quarterly Board meetings, scheduled at a regular permanent time throughout the

project, enable the Board to plan ahead throughout the entire project for meetings and also allow for frequent timely input on all stages of the research.

To maximize the effectiveness of the Board, the PIs will prepare and distribute materials prior to each meeting, including summaries of work to date, proposals for details of research plans or measures, drafts of papers and presentations, etc. Detailed notes will be taken during each meeting, and the meetings will also be recorded for post-hoc review. Members will also be invited to post notes, comments, and links in the meeting chats which will also be recorded. All materials from the meetings, including the slide shows, the written agenda and links to documents, and the recordings will be made available to Board members to review after the meetings.

PI Warschauer will be responsible for managing communications with the Board, drafting documents for the Advisory Board (which will be shared with co-PIs for their approval), and ensuring that the Board members' suggestions are implemented. In particular, Warschauer will maintain a checklist of suggestions made by the Board and keep track and report back as to how the project team responded to suggestions.

Individual Board members will also be consulted outside the scheduled meetings on issues related to their expertise. For example, **Kalinowski** and **Melzi** will provide guidance on approaches to developing effective narratives for children's science learning, and will review and give feedback on the stories that are developed. **Lovato** and **Tare** will consult about effective ways to reach national audiences with informal learning content and **Aviles** will assist us in developing and implementing our national strategy for Latine community outreach, leveraging her extensive networks in these communities.

F. PROJECT MANAGEMENT

Mark Warschauer (PI) is a Professor of Education, Informatics, Language Science, and Psychological Science at UCI, where he directs the Digital Learning Lab, and a member of the National Academy of Education. He has served as PI of a wide range of projects related to digital learning, science learning, natural language processing, and learning among culturally and linguistically diverse learners. He will supervise the overall project; lead the research; and maintain relations with research partners, sites, and the Advisory Board.

Andres Bustamante (Co-PI) is Assistant Professor of Education at UCI with expertise in young Latine children's science learning in informal settings, and served as the lead content expert for the California Department of Education committee that developed the new early science learning standards. He is currently PI of a large NSF grant codesigning and evaluating informal science learning environments in partnership with Latine families in SAELI. Bustamante will provide leadership in areas related to co-design with Latine families and in science content and assessment.

Ying Xu (Co-PI) is Assistant Professor of Learning Sciences and Technology at University of Michigan where she specializes in developing AI-based children's media to promote dialogic science learning. She will provide expertise in development of the e-book platform and in lab- and home-based experimental design.

Teresa Satterfield (Co-PI) is Professor of Spanish at UM and founder and director of the ENL Saturday community program for Spanish-speaking children. She will provide expertise on child bilingualism and biliteracy and on Latine identity and culture and assist with coordinating work at ENL.

Stephanie Soto-Lara (Co-PI) is Project Director of the Santa Ana Early Learning Initiative. She will provide expertise in culturally-sustaining STEM education and family support for children's science learning, and will help coordinate co-design, research, and dissemination activities in California.

F.1 Collaboration

Warschauer, Bustamante, Xu, and Soto-Lara have worked together extensively, including on previous NSF-funded projects, and Xu and Satterfield have also collaborated previously. Bustamante and Satterfield will take the lead on developing co-design activities, and Warschauer and Xu on development of the dialogic e-books. They will be assisted in these tasks by two graduate student researchers (one at UCI and one at UM) and a postbac research assistant (at UCI). The project leaders and three research staff will meet weekly via videoconference to coordinate their work. They will additionally be assisted by a part-time staff at UCI with responsibilities for software engineering.